Harshal Parekh

+91 8238482880 | harshalparekh17@gmail.com | linkedin.com/in/harshal-p-parekh | github.com/harshal4172005 | Portfolio
Surat, India | Open to relocation | Available to join immediately

Summary

2026 Computer Engineering graduate with 8 months of industry experience as an AI Engineer at NexAI Labs, where I built and deployed AI image and video pipelines for 5+ brands with FastAPI and Docker. Also shipped a full-stack multi-branch ERP (.NET, React, PostgreSQL) using AI coding agents (Claude Code, Cursor, Antigravity). Seeking software, AI/ML and backend roles..

Experience

Junior AI Engineer Nov. 2025 – Apr. 2026
NexAI Labs *Surat, India*

- Built diffusion-model image generation pipelines for 5+ garment and jewellery brands, cutting manual design effort by 60%.
- Automated image generation and content creation end-to-end with n8n multi-agent workflows, unifying model APIs.
- Extended the pipeline to AI video generation, speeding up client content delivery by 40%.
- Deployed the AI services as REST APIs (FastAPI, Flask, Docker) on AWS EC2, storing media in S3 and Cloudflare R2.

AI Engineer Intern Aug. 2025 – Oct. 2025
NexAI Labs *Surat, India*

- Built prompt workflows and API integrations for AI automation pipelines, with Pandas/NumPy data processing.
- Deployed and tested FastAPI-based ML applications on Google Cloud.

Projects

Multi-Branch Gaming Cafe ERP | *C#, .NET 8, React, Tailwind CSS, PostgreSQL, SignalR, Redis, Docker* GitHub

- Built and deployed a full-stack ERP for gaming cafés, designed to manage multiple branches and 100+ gaming PCs.
- Built an ASP.NET Core (C#) API (Clean Architecture, 30+ controllers, EF Core, Swagger) and a 16-module React UI.
- Synced sessions, billing and orders live via 10 SignalR hubs, plus a Windows agent that locks and unlocks PCs remotely.
- Automated usage-based billing with split payments (cash, UPI, wallet), member wallets and tamper-proof daily reports.
- Added JWT role-based auth; deployed 7 Docker services on an Ubuntu server (Oracle Cloud) with Nginx, SSL, backups.
- Built with AI coding agents (Claude Code, Cursor, Antigravity): wrote specs, scoped tasks, reviewed and debugged their code.

Multimodal RAG Research Assistant | *Python, LangChain, FastAPI, Pydantic, Gemini, Qdrant, Docker* GitHub

- Built a multimodal RAG assistant with LangChain and Qdrant over 7 AI papers, with page-cited answers via FastAPI.
- Captioned every figure with Gemini Vision, incl. vector diagrams most PDF tools miss, so figures are searchable in Qdrant.
- Added a Gemini, Groq and NVIDIA NIM fallback chain with backoff retries so rate limits never block answers.

AI CCTV Safety Surveillance | *Python, YOLOv8, PyTorch, OpenCV, Pandas, NumPy, Streamlit, SQLite* GitHub

- Fine-tuned YOLOv8 on a 10-class PPE dataset to detect missing hard hats, masks and vests in images and live video.
- Logged every violation with a timestamp and cropped snapshot, with a per-class cooldown to stop duplicate alerts.
- Built a Streamlit app with live WebRTC video, Plotly analytics, SQLite login and Pandas-based PDF/CSV reports.

Technical Skills

Languages: Python, SQL, Java (basic)
AI / ML: Diffusion Models, YOLOv8, OpenCV, PyTorch, Pandas, NumPy, LangChain, RAG, Qdrant, Gemini API
AI Dev Tools: Claude Code, Cursor, GitHub Copilot, OpenAI Codex, Antigravity, Augment Code
Automation: n8n, Multi-Agent Workflows, API Integrations
Backend & APIs: FastAPI, Flask, ASP.NET Core, EF Core, SignalR, REST APIs, JSON, Pydantic, JWT, Swagger
DevOps & Cloud: Docker, Nginx, Linux, Git, GitHub, AWS (EC2, S3), Google Cloud, Oracle Cloud, Cloudflare R2
Databases: PostgreSQL, SQLite, Redis
Frontend & Tools: React, Tailwind CSS, HTML, CSS, Streamlit

Education

C.K. Pithawala College of Engineering and Technology Surat, India
B.E. in Computer Engineering – CGPA: 8.41 / 10 *Aug. 2022 – May 2026*

Certifications

Ethical Hacking & Penetration Testing – C-DAC Noida (training + virtual internship)
AWS Cloud Practitioner (CLF-C02) Exam Prep – Udemy
Python, Data Analysis & AI Foundation – SAP Code Unnati (Edunet Foundation)
Google Cloud Skills Boost (22 badges) – Gen AI with Gemini, Cloud Run, Vision API, Prompt Design (Profile)